Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	29 September 2026
Team ID	SWTID-2026-3075
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders
Team Size	4
Team Leader	Karpagam R
Team Members	R.Padma Priya, S.Mariyam Asiya,S.Shreya
Maximum Marks	4 Marks

Technical Architecture:

Project objective: Automate the standard laptop procurement workflow in ServiceNow University. The application allows users to submit a standard laptop request, validates the request, routes it through approval, creates/updates procurement records, and sends status notifications with minimal manual effort.

Architecture: User → ServiceNow Service Catalog → Flow Designer / Business Rules → Approval & Procurement Logic → ServiceNow Data Tables → Notifications / Reports. The solution is deployed on the ServiceNow cloud platform.

USER Employee / Requester	**SERVICENOW SERVICE CATALOG** Standard Laptop Request Form	**AUTOMATION LAYER** Flow Designer • Business Rules • Script Includes
APPROVAL Manager / IT Procurement	**DATA LAYER** Request • Catalog Task • Asset / Procurement Records	**OUTPUT** Notifications • Status • Reports / Dashboard

Technology Stack / Architecture Components

S.No.	Component	Description	Technology
1	**User Interface**	Employee-facing laptop request form and Service Catalog experience.	ServiceNow Service Catalog, Catalog Item, Service Portal / Workspace
2	**Application Logic-1**	Workflow logic for request validation, routing and task creation.	Flow Designer
3	**Application Logic-2**	Server-side logic for validation and automation where required.	Business Rules, Script Includes, Glide APIs
4	**Application Logic-3**	Approval and procurement automation.	Approval Rules, Flow Designer
5	**Database**	Stores requests, catalog tasks, users, approvals and procurement-related records.	ServiceNow platform database / application tables
6	**Cloud Database**	Managed cloud data layer provided by the ServiceNow platform.	ServiceNow Cloud Platform
7	**File Storage**	Optional storage for procurement documents or supporting attachments.	ServiceNow Attachments / platform file storage

Technology Stack (Architecture & Stack) — Continued

S.No.	Component	Description	Technology
8	**External API-1**	No mandatory third-party API is required for the core workflow; integration can be added for procurement/vendor systems.	ServiceNow REST API / IntegrationHub, if required
9	**External API-2**	Optional integration with organization procurement, inventory or email systems.	REST / SOAP APIs, IntegrationHub
10	**Machine Learning Model**	Not required for the standard laptop ordering automation workflow.	Not applicable
11	**Infrastructure (Server / Cloud)**	Application is hosted and operated on the ServiceNow cloud platform; no local application server is required.	ServiceNow Cloud / SaaS platform

Table-2: Application Characteristics

S.No.	Characteristics	Description	Technology
1	**Open-Source Frameworks**	The core solution uses the ServiceNow platform rather than a separate open-source application framework. JavaScript is used for supported platform scripting.	JavaScript / ServiceNow APIs
2	**Security Implementations**	Role-based access, ACLs, authentication, authorization and controlled access to procurement records.	ServiceNow ACLs, Roles, Authentication, HTTPS/TLS
3	**Scalable Architecture**	Cloud-based SaaS architecture supports multiple users and automated workflow execution without maintaining local servers.	ServiceNow Cloud Platform, Flow Designer
4	**Availability**	The application runs on the ServiceNow cloud service, reducing dependency on local infrastructure and enabling centralized access.	ServiceNow Cloud / SaaS

Core Automated Flow

1. Request	Employee selects the standard laptop catalog item and submits required details.
2. Validation	Mandatory fields and request conditions are checked automatically.
3. Approval	Request is routed to the configured manager / IT procurement approver.
4. Procurement Task	Approved requests generate procurement / fulfillment tasks for the IT team.
5. Notification	Requester receives status updates through ServiceNow notifications.
6. Closure	Fulfillment is recorded and the request is closed after completion.

Technology Stack (Architecture & Stack) — Application Performance

S.No.	Characteristics	Description	Technology
5	**Performance**	Performance is improved by automating repetitive approval and task-creation steps, minimizing manual processing, using efficient server-side logic, and avoiding unnecessary client-side processing. The design is suitable for concurrent users within the ServiceNow platform.	Flow Designer, Business Rules, Glide APIs, ServiceNow Cloud

Project Team

Role	Name	Team ID
Team Leader	Karpagam R	SWTID-2026-3075
Team Member	Padma Priya R	SWTID-2026-3075
Team Member	Mariyam Asiya S, Shreya S	SWTID -2026-3075

References

• ServiceNow University — ServiceNow learning platform and project environment.

• ServiceNow Product Documentation — Service Catalog, Flow Designer, Business Rules, ACLs and platform architecture.

• ServiceNow Developer Documentation — JavaScript APIs, Glide APIs and application development.

Note: The architecture is tailored to the project scope “Streamlining IT Procurement: Automating Standard Laptop Orders.” Optional external integrations are shown as extensibility points; the core workflow can be implemented entirely within ServiceNow.